

Richard Liu

guanzhi@uchicago.edu | [Github](#) | [Website](#)

Education

University of Chicago

4th year PhD Student, Computer Science
Advisor: Rana Hanocka

2022 – Expected 2027

University of Chicago

B.A. Economics, B.S. Mathematics (GPA 3.73)

2015 – 2019

Publications

Research Interests: 3D graphics and geometry processing from 2D priors, 3D generation

2D Features Are All You Need for 3D Shape Understanding

2026

Jinfan Zhou, **Richard Liu***, Itai Lang, Rana Hanocka
ECCV 2026 ([Website](#))

Deep Feature Deformation Weights

2026

Richard Liu, Itai Lang, Rana Hanocka
CVPR 2026 Highlight ([Paper](#), [Website](#), [Code](#), [Video](#))

WIR3D: Semantic and Geometric-Aware 3D Shape Abstraction

2025

Richard Liu, Daniel Fu, Noah Tan, Itai Lang, Rana Hanocka
ICCV 2025 Oral ([Paper](#), [Website](#), [Code](#))

TEDi: Temporally-Entangled Diffusion for Long-Term Motion Synthesis

2024

Jack Zhang, **Richard Liu**, Noam Aigerman, Vladimir G. Kim, Rana Hanocka
SIGGRAPH North America 2024 ([Paper](#), [Website](#), [Code](#))

Towards Multimodal Interaction with AI-Infused Shape-Changing Interfaces

2024

Chenfeng Gao, Wanli Qian, **Richard Liu**, Rana Hanocka, Ken Nakagaki
UIST Poster 2024 ([Paper](#), [Video](#))

HyperFields: Towards Zero-Shot Generation of NeRFs from Text

2024

Sudarshan Babu*, **Richard Liu***, Avery Zhou*, Michael Maire, Gregory Shakhnarovich, Rana Hanocka
ICML 2024 ([Paper](#), [Website](#), [Code](#))

DA Wand: Distortion-Aware Selection using Neural Mesh Parameterization

2022

Richard Liu, Noam Aigerman, Vladimir G. Kim, Rana Hanocka
CVPR 2023 ([Paper](#), [Website](#), [Code](#), [Talk](#))

Text2Mesh: Text-Driven Neural Stylization for Meshes

2021

Oscar Michel*, Roi Bar-On*, **Richard Liu***, Sagie Benaim, Rana Hanocka
CVPR 2022 Oral ([Paper](#), [Website](#), [Code](#))

Designing Online Advertisements via Bandit and Reinforcement Learning

2021

Richard Liu, Yusuke Narita, Kohei Yata
(Peer-Reviewed) ICML 2021 RL4RealLife Workshop ([Paper](#))

Invited Talks

Symposium on Geometry Processing

2026

“Geometry Processing from 2D Image Priors”
University of Bern ([Recording](#), [Slides](#))

Summer Geometry Institute

2024-2026

“Introduction to Mesh Parameterization”
MIT ([Recording](#), [Slides](#))

Geometric Computation Group (Guibas Lab) “Neural Mesh Parameterization: DA-Wand and Beyond” Stanford (Slides)	2023
--	------

CMSC 23700: Introduction to Computer Graphics “Introduction to Conformal Mesh Parameterization” University of Chicago	2022-2026
--	-----------

Professional Activities

Workshop Organizer Great Lakes Graphics University of Chicago (Recording)	2025
--	------

Reviewer CVPR (2023-26), SIGGRAPH North America (2023-26), SIGGRAPH Asia (2023-26), ICCV/ECCV (2023-26), Eurographics (2023), 3DV (2025-26), CAD (2023, 2024), CGF (2022), TVCG (2025)	
--	--

DA Wand Blender Extension Implemented and released DA Wand trained model as a Blender extension (extension link)	2025
---	------

SIG Project Leader Summer Geometry Institute	2022
--	------

Replicability Stamp Reviewer (GRSI)	2022, 2025
--	------------

Teaching Assistant CMSC 23700: Introduction to Computer Graphics	2022-2023
--	-----------

Awards

Eckhardt Graduate Fellowship	2022 - Ongoing
ITA Tech Challenge	Top 150
Dean's List	2015-2019
University Scholar Fellowship	2015-2019

Work Experience

Founding Research Engineer Thrixel	2026-Present
--	--------------

University of Chicago 3DL Rana Hanocka PhD Student	2022 - Present
---	----------------

PhD Intern Adobe Research	Summer 2023
-------------------------------------	-------------

Yale Tobin Center for Economic Policy John Eric Humphries and Yusuke Narita Predoctoral Fellow	2020 - 2021
---	-------------

Booth School of Business Anita Rao Research Professional	2020 – 2020
---	-------------

Voleon Capital Management Operations Associate	2019 – 2020
--	-------------